

Introduction to the Structure of Braid Groups

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Abstract

This short document serves as an introduction to the topic of a *braid group on n strings* B_n , which is a certain type of group that appears in various contexts in mathematics, but which also has its place in different applications. The topics in this document are predominantly based on chapters 1.1 through 1.4 of the book “Braid Groups” by Christian Kassel and Vladimir Turaev [CK08]. We first talk about three different approaches for viewing braid groups: Purely algebraic, geometrically and as fundamental groups arising in the context of algebraic topology. Afterwards, the focus lies on structural properties of the braid groups, especially for specific subgroups called *pure braids*, where the main goal is to see that $B_n \not\cong B_m$ for $n \neq m$. Along this journey, we will see that pure braids admit a *normal form*. Furthermore, some interesting connections to other mathematical areas, such as homotopy theory and symmetric groups will be explored. In particular, results about fibrations and induced exact sequences can be used to find generators for pure braid groups. This is a case where algebraic topology helps for solving an algebraic problem, where oftentimes it is the other way round.

Contents

In chapter 1, a motivational example for braid groups coming from the area of topological robotics is given. In chapter 2, some definitions and results, which may be unfamiliar to the reader, from algebra, group theory and algebraic topology are established. In chapter 3, different versions of braid groups are defined, which turn out to be all isomorphic to each other. Furthermore, connections to symmetric groups are established in this chapter. Following that, chapter 4 serves as the main part of this document. First, some fundamental homomorphisms as well as the normal form for pure braids are considered. Afterwards, generators for pure braid groups are discussed. Then it will be shown that pure braid groups are torsion-free. All of these results are used in the last part of this chapter which is about the center of braid groups, from which the main result can be deduced, namely that $B_n \not\cong B_m$ for $n \neq m$.

1 Introduction

Imagine the following scenario: You are working at the factory planning department of some big company and someone comes in at monday morning and instructs you to analyze motion patterns for the new fancy autonomous robots they bought recently. You ask yourself: How to model this in mathematical terms? Before tackling this question, one can make some simplifying assumptions: The factory ground is assumed to be represented by an unbounded rectangle without any obstructions (i.e. $\mathbb{R}^2$), while the robots are assumed to have no volume, so they can be represented by single points in $\mathbb{R}^2$. Of course, two robots cannot occupy exactly the same space, so we must assume that the chosen points in $\mathbb{R}^2$ are pairwise different. For n robots, this modelling process leads us to the *configuration space of n points in $\mathbb{R}^2$*

$$\mathcal{F}_n(\mathbb{R}^2) = \{(x_1, \dots, x_n) \in (\mathbb{R}^2)^n \mid x_i \neq x_j \text{ for } i \neq j\}.$$

A point in this space then represents a choice of placement on the factory floor for each robot. Now all of the robots should execute some tasks involving movement, so we assume that they move around the factory simultaneously in some given time frame. For simplicity's sake, we assume that the time frame is given by the unit interval $I = [0, 1]$. Furthermore, it makes sense to assume that the movement must be somehow continuous, since the robots are not allowed to break the laws of physics via teleportation. Coming back to our configuration space model, by equipping I with the standard topology induced by $\mathbb{R}$ and considering $\mathcal{F}_n(\mathbb{R}^2)$ as a subspace of $(\mathbb{R}^2)^n$ with the product topology, this means we have to look at continuous paths

$$I \rightarrow \mathcal{F}_n(\mathbb{R}^2).$$

Let us simplify even more and consider only those paths which are loops, i.e. those movements, where start- and end-point are the same for each robot. If we additionally do not care about exact distances between the robots (as long as they do not occupy the same space, which would result in a crash), which amounts to allowing continuous deformations of the individual robot paths, we arrive at the *fundamental group*

$$\pi_1(\mathcal{F}_n(\mathbb{R}^2))$$

of the configuration space. The elements of this group are equivalence classes of homotopic loops with multiplication given by concatenation of paths.

Before continuing further, let's talk about visualization. Using

$$\mathcal{F}_n(\mathbb{R}^2) \subseteq \underbrace{\mathbb{R}^2 \times \dots \times \mathbb{R}^2}_{n \text{ times}},$$

by properties of the product topology, a loop γ in this space is the same as a collection of n loops $\gamma_1, \dots, \gamma_n$ in

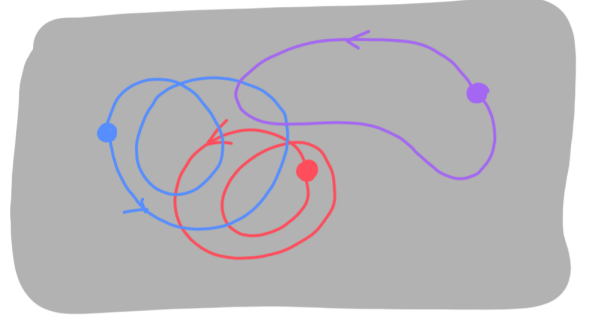


Figure 1: Naive visualization approach for a loop in $\mathcal{F}_3(\mathbb{R}^2)$.

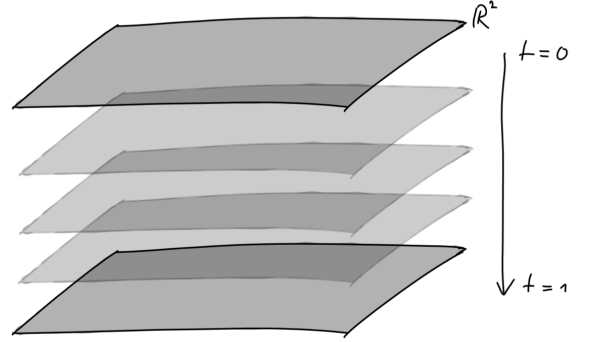


Figure 2: Stacking copies of $\mathbb{R}^2$ on top of each other, where each layer corresponds to a time t .

$\mathbb{R}^2$, such that $\gamma_i(t) \neq \gamma_j(t)$ for $i \neq j$ at any given time t . Then γ could be visualized as in fig. 1, but the problem here is, that the information about the placement of the individual robots at a given time t is lost completely. However, this information loss is crucial, since the robots are not allowed to crash into each other.

Thus, we adopt a different approach: Imagine, that for each $t \in I$ we have a copy of $\mathbb{R}^2$. Now we “stack” those copies on top of each other, such that from top to bottom the individual “layers” correspond to t ranging from 0 to 1, as depicted in fig. 2. Using this approach, the loop from fig. 1 could be visualized (up to homotopy) as in fig. 3.

The observant reader might already see where this is going: Fig. 3 suggests that, looking from the “front”, there are some kinds of “crossings” where one path goes “under” the other in such a way that is reminiscent of (real-world) braids. Indeed, loops of this kind will be the main subject of this document, even though the connection between algebraic/geometric braids and fundamental groups of configuration spaces has to wait until section 3.4. For the rest of this introduction, let us de-

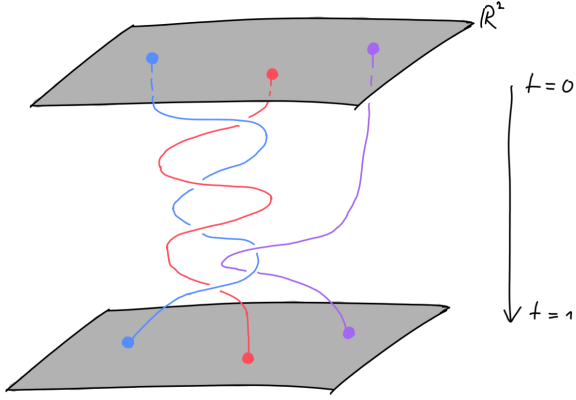


Figure 3: Better visualization of a loop in $\mathcal{F}_n(\mathbb{R}^2)$.

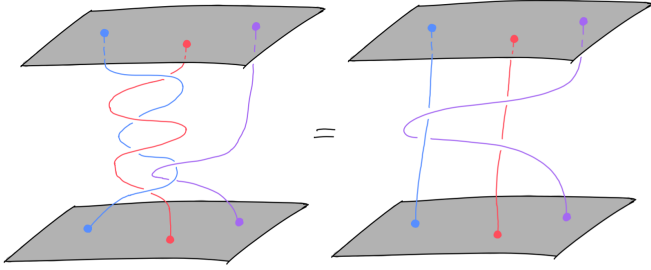


Figure 4: Two braids that are the same up to homotopy.

note the elements of $\pi_1(\mathcal{F}_n(\mathbb{R}^2))$ as *braids*.

Having the visualizations in mind, several questions could arise. When are two braids essentially the same? In this context, with “essentially” we mean “up to homotopy”. For example, it turns out that the braids in fig. 4 are essentially the same, which can be seen by pulling the strings around without allowing the strings to cross through each other. Another question one could ask is: What are the basic building blocks for these braids? Or more generally, what structural properties can be discovered? This is the motivation for what will come in the next chapters.

2 Preliminaries

Before diving into the main topic, we need to clarify some preliminaries in order to ensure that we are on the same page. Basic knowledge about groups as well as some knowledge about algebraic topology and fundamental groups is assumed, but we will elaborate on more specific topics that some readers might not be acquainted with.

2.1 Notation

By *homomorphism* we always mean group homomorphism, unless otherwise noted.

Given a homomorphism $f : G \rightarrow H$, we write $G \xhookrightarrow{f} H$ if f is injective, $G \twoheadrightarrow H$ if f is surjective, and $G \dashrightarrow H$ if the existence of f is implied by something else.

The trivial group is denoted as 1. We write $H \leq G$ if H is a subgroup of G . In this case $H \hookrightarrow G$ denotes the inclusion homomorphism. Isomorphic groups G and H are written as $G \cong H$. In the context of topological spaces, homotopy-equivalent spaces are written as $X \simeq Y$.

2.2 Exact Sequences

A sequence

$$\dots \xrightarrow{f_{i-2}} G_{i-1} \xrightarrow{f_{i-1}} G_i \xrightarrow{f_i} G_{i+1} \xrightarrow{f_{i+1}} \dots$$

of groups and homomorphisms between them is called *exact at G_i* , if $\ker f_i = \operatorname{im} f_{i-1}$. We call the whole sequence *exact*, if it is exact at each index. A *short exact sequence* is an exact sequence of the form

$$1 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 1$$

where 1 is the trivial group. All other exact sequences not of this form are called *long exact sequences*.

Note that for a short exact sequence we always have $\ker f = \operatorname{im}(1 \rightarrow A) = 1$ and $\operatorname{im} g = \ker(C \rightarrow 1) = C$, implying that f is injective and g is surjective.

There are some noteworthy special cases of short exact sequences:

- If $A \cong 1$, then g is an isomorphism.
- If $C \cong 1$, then f is an isomorphism.
- If N is a normal subgroup of G , then

$$1 \longrightarrow N \hookrightarrow G \twoheadrightarrow G/N \longrightarrow 1$$

is a short exact sequence.

Whenever one has a short exact sequence as defined above, it always holds that

$$C \cong B/f(A).$$

A *split exact sequence* is a short exact sequence of the form above, such that there is a homomorphism $h : C \rightarrow B$ such that $g \circ h = \operatorname{id}_C$ is the identity on C :

$$1 \longrightarrow A \xrightarrow{f} B \begin{matrix} \xrightarrow{g} \\ \nwarrow h \end{matrix} C \longrightarrow 1$$

2.3 Group Theory

The results in this chapter can be found in introductory works to group theory, for example [Lan02], chapter I.

Given a group G , elements $g_1, g_2 \in G$ are called *conjugate*, if there is $h \in G$ such that $g_2 = hgh^{-1}$. A subgroup $N \leq G$ is called *normal*, if it is invariant under conjugation, i.e. we have $gNg^{-1} \subseteq N$ for all $g \in G$.

Given a group homomorphism $f : G \rightarrow H$, denote its image as $\text{im } f$ and its kernel as $\ker f$. Note that $\ker f$ is a normal subgroup of G .

If N is a normal subgroup of G , the *quotient group*

$$G/N = \{aN \mid a \in G\}$$

carries a group structure given by

$$(aN) \cdot (bN) = (ab)N.$$

This group is characterised by the following universal property: Given a homomorphism $f : G \rightarrow H$ such that $N \subseteq \ker f$, there is a unique homomorphism $\bar{f} : G/N \rightarrow H$ making the following diagram commute:

$$\begin{array}{ccc} G & \xrightarrow{f} & H \\ \downarrow & \nearrow \exists! \bar{f} & \\ G/N & & \end{array}$$

Here, the vertical map is the canonical projection onto the quotient group.

Since $\ker f$ is always a normal subgroup, we know that $\text{coim } f = G/\ker f$ has a well-defined group structure.

Given $g, h \in G$, the *commutator* $[g, h]$ is $g^{-1}h^{-1}gh$ and the *commutator subgroup* of G is $[G, G] = \langle [g, h] \mid g, h \in G \rangle$. It is a normal subgroup. Note that G/N is abelian if and only if $[G, G] \subseteq N$.

The *abelianization* G^{ab} of G is $G^{\text{ab}} := G/[G, G]$. Induced by the universal property of the quotient, it has the following universal property: Given a homomorphism f from G to an abelian group H , there is a unique homomorphism $\bar{f}$ making the following diagram commute:

$$\begin{array}{ccc} G & \xrightarrow{f} & H \\ \downarrow & \nearrow \exists! \bar{f} & \\ G^{\text{ab}} & & \end{array}$$

The *free group* F_S generated by a set S has the following universal property: Given any map f from S to a group G , there is a unique homomorphism $\varphi : F_S \rightarrow G$,

making the following diagram commute:

$$\begin{array}{ccc} S & \hookrightarrow & F_S \\ & \searrow f & \downarrow \exists! \varphi \\ & & G \end{array}$$

Given a group G , a subgroup $H \leq G$ and a normal subgroup $N \leq G$, we call G the *semidirect product* of N and H , written as $G = N \rtimes H$, if there is a split exact sequence

$$1 \longrightarrow N \longrightarrow G \longrightarrow H \longrightarrow 1.$$

In this case, G is equal to $N \cdot H$ with $N \cap H = \{1\}$.

2.4 Algebraic Topology

The following definition along with their corresponding results can be found in [Hat02] chapter 4.1.

Definition 2.1. Given a topological space X with base-point x_0 , define $\pi_n(X, x_0)$ to be the set of homotopy classes of maps $g : [0, 1]^n \rightarrow X$ with the property that $g(\partial([0, 1]^n)) = \{x_0\}$.

This can be given the structure of an abelian group for $n \geq 2$. For $n \geq 1$ it is equal to the fundamental group of a space. For $n = 0$, the set corresponds to path-components of X .

Given topological spaces X, Y with basepoints $x_0 \in X, y_0 \in Y$, a map of pairs $f : (X, x_0) \rightarrow (Y, y_0)$ is a continuous map $f : X \rightarrow Y$ such that $f(x_0) = y_0$. For $n \geq 1$, such a map induces a homomorphism

$$f_{\#} : \pi_n(X, x_0) \rightarrow \pi_n(Y, y_0), \quad [\gamma] \mapsto [f \circ \gamma].$$

Definition 2.2. Given topological spaces F, E, B , a *locally trivial fibration with fiber F* is a continuous map $p : E \rightarrow B$ such that for every $x \in B$ there is a neighborhood $U \subseteq B$ and a homeomorphism

$$\varphi : U \times F \rightarrow p^{-1}(U)$$

making the following diagram commute:

$$\begin{array}{ccc} U \times F & \xrightarrow[\cong]{\varphi} & p^{-1}(U) \\ & \searrow p_1 \quad \swarrow p & \\ & U & \end{array}$$

Here, $p_1 : U \times F \rightarrow U$ denotes the projection onto the first factor. We call F the *fiber*, E the *total space* and B the *base space* of p .

In other words, a locally trivial fibration is a topological space E with added structure, namely that it can be thought of as being “parametrized” by B , in the sense that locally (with respect to B) it behaves just like the

trivial fibration $B \times F$. From the definition it follows that $F \cong p^{-1}(b)$ for every $b \in B$.

A fibration as given above is often written as

$$F \hookrightarrow E \twoheadrightarrow B$$

Without going into further detail, every locally trivial fibration is a *Serre fibration*. An important result in homotopy theory is that Serre fibrations induce specific long exact sequences. For more details on this topic, see for example [DBF84], chapter 5.

Proposition 2.3 (Homotopy Sequence). *Let $p : E \rightarrow B$ be a Serre fibration. Pick a basepoint $e \in E$, set $b := p(e)$ and denote $F := p^{-1}(b)$. Consider*

$$F \xhookrightarrow{\iota} E \xrightarrow{p} B$$

Then there is a long exact sequence

$$\begin{array}{ccccccc} & & & & \partial & \longrightarrow & \dots \\ & & & & \swarrow & & \\ \pi_2(F, e) & \xleftarrow{\iota_{\#}} & \pi_2(E, e) & \xrightarrow{p_{\#}} & \pi_2(B, b) & & \\ & & \swarrow & & \partial & \longrightarrow & \\ \pi_1(F, e) & \xleftarrow{\iota_{\#}} & \pi_1(E, e) & \xrightarrow{p_{\#}} & \pi_1(B, b) & & \\ & & \swarrow & & \partial & \longrightarrow & \\ \pi_0(F, e) & \xleftarrow{\iota_{\#}} & \pi_0(E, e) & \xrightarrow{p_{\#}} & \pi_0(B, b) & & \end{array}$$

where $\iota_{\#}$ and $p_{\#}$ denote the induced maps on homotopy groups and ∂ is a map constructed using properties of Serre fibrations.

Exactness for the last three arrows means $\text{im } \partial = \ker \iota_{\#}$ and $\text{im } \iota_{\#} = \ker p_{\#}$, where in this case kernel means preimage of the distinguished element.

3 Different Views of Braid Groups

3.1 Through the Lens of Geometry

In the introduction we have seen that “braids” arise naturally in the context of topological motion planning. But before discussing the connection to algebraic topology in more detail, we first consider braids as purely geometric objects. It will turn out in the next chapter, that geometric braids are isomorphic to *Artin Braid Groups*, which pose a way to study braids in a purely algebraic way.

We do not go into much detail here regarding the proofs and properties of geometric braids, since we use them mainly as a tool for visualization and as a justification for showcasing algebraic properties via pictures.

It should be noted, however, that geometric braids are interesting objects in their own right and have connections to other areas, such as knot theory, see for example

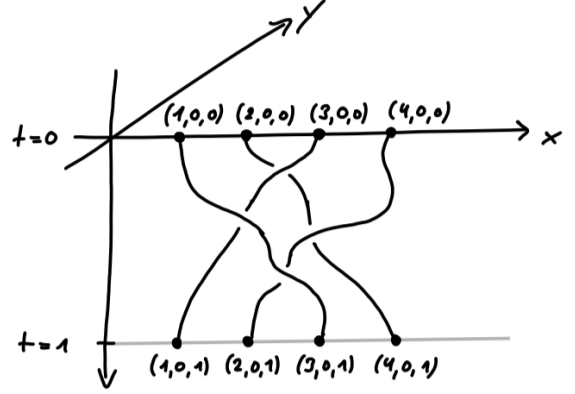


Figure 5: Example of a geometric braid.

theorem 2.3 in [CK08].

Definition 3.1. A geometric braid on $n \geq 1$ strings is a set $b \subseteq \mathbb{R}^2 \times I$ formed by n disjoint topological intervals, such that

- i) The projection $\mathbb{R}^2 \times I \rightarrow I$ maps each string homeomorphically onto I ,
- ii) $b \cap (\mathbb{R}^2 \times \{0\}) = \{(1,0,0), \dots, (n,0,0)\}$,
- iii) $b \cap (\mathbb{R}^2 \times \{1\}) = \{(1,0,1), \dots, (n,0,1)\}$.

By i) it is then clear, that each string meets each plane $\mathbb{R}^2 \times \{t\}$ in exactly one point. Together with conditions ii) and iii) we have that each string connects a point $(i,0,0)$ to some other point $(s(i),0,1)$, where $i, s(i) \in \{1, \dots, n\}$, such that $i \mapsto s(i)$ is bijective. We call this permutation $(s(1), \dots, s(n))$ of $\{1, \dots, n\}$ the *underlying permutation* of a braid b .

Example 3.2. i) An example for a braid is given in fig. 5.

ii) The string in fig. 6 satisfies conditions ii) and iii), but condition i) fails. The string is not homeomorphic to I , since locally around the crossing, the string does not behave like I (topologically speaking). Thus it is not a braid.

iii) The string in fig. 7 again violates condition i), since there is a point in I that has multiple preimages under the projection $\mathbb{R}^2 \times I \rightarrow I$. Thus it is also not a braid.

Two braids b, b' on n strings are called *isotopic*, if there is a continuous map $F : b \times I \rightarrow \mathbb{R}^2 \times I$, such that for each $s \in I$ the map

$$F_s : b \rightarrow \mathbb{R}^2 \times I, \quad x \mapsto F(x, s)$$

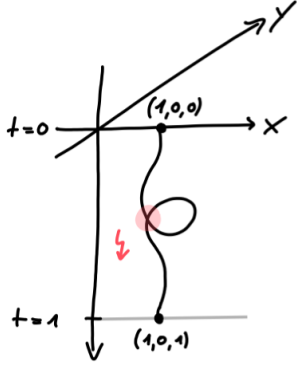


Figure 6: Not a geometric braid.

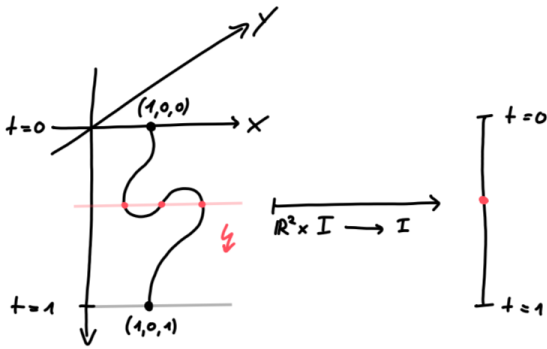


Figure 7: Also not a geometric braid.

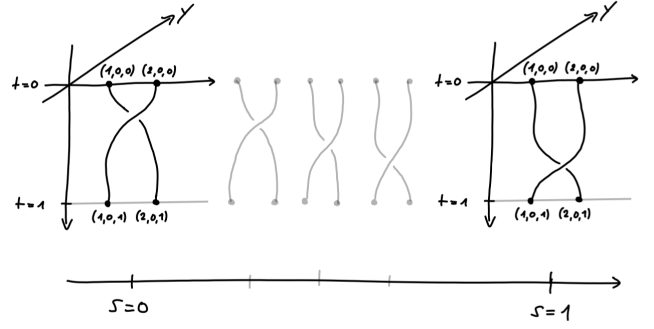


Figure 8: An isotopy between two braids.

is an embedding whose image is a geometric braid on n strings and such that the following diagram commutes:

$$\begin{array}{ccc}
 b \times \{0\} & \xrightarrow{p_1} & b \\
 \downarrow & & \downarrow \\
 b \times I & \xrightarrow{F} & \mathbb{R}^2 \times I \\
 \uparrow & & \uparrow \\
 b \times \{1\} & \xrightarrow{F} & b'
 \end{array}$$

Here p_1 is again the projection onto the first factor and the bottom arrow is required to be surjective.

In particular, this means that $F_0 : b \rightarrow b$ is the identity and F_1 maps b surjectively onto b' . Intuitively, an isotopy between two braids b, b' can be understood as a family of braids that form a continuous “interpolation” between b and b' , see fig. 8. Notably, through this “continuous deformation”, it is not allowed that one string “crosses through” another string, since this would violate the condition that at each “interpolation index” s , the image of s is again a geometric braid.

In order to get a group structure on the set of isotopy classes of braids, define the product of two n -string geometric braids b_1, b_2 to be the set of points $(x, y, t) \in \mathbb{R}^2 \times I$ such that $(x, y, 2t) \in b_1$ for $0 \leq t \leq 1/2$ and $(x, y, 2t - 1) \in b_2$ for $1/2 \leq t \leq 1$, see fig. 9.

The neutral element with respect to this multiplication is the trivial braid 1 represented by the geometric braid

$$\{1, 2, \dots, n\} \times \{0\} \times I,$$

see fig. 10.

Without going into further detail (for more info, see [CK08] chapter 1.2.1), this defines a group structure on the set of isotopy classes of geometric braids.

Definition 3.3. We define this set with the given group structure as *geometric braids on n strings* $\mathcal{B}_n$.

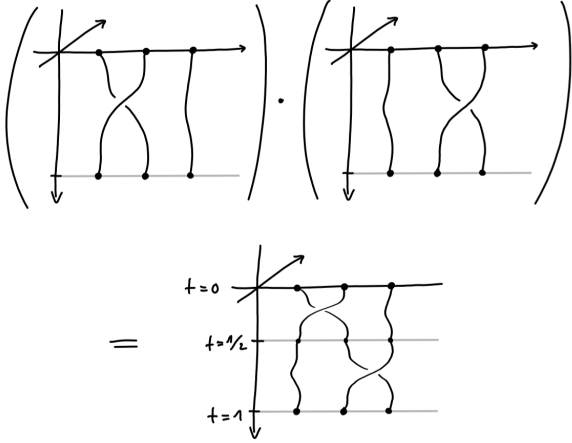


Figure 9: The product of two braids.

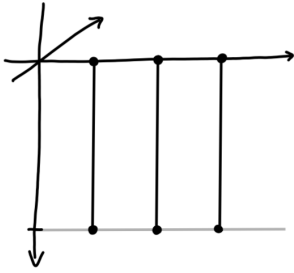


Figure 10: The identity braid on 3 strings.

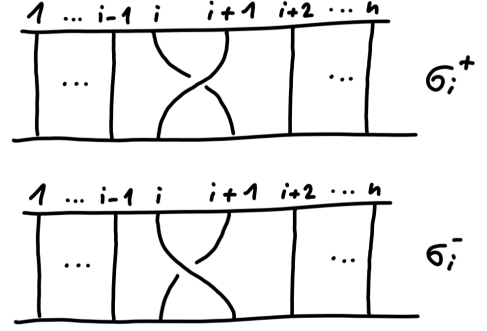


Figure 11: The elementary (geometric) braids σ_i^+ and σ_i^- .

Our main tool for visualization will be *braid diagrams*. A braid diagram of a given braid $b \in \mathcal{B}_n$ can be understood as the projection of b onto $\mathbb{R} \times \{0\} \times I$, while tracking the information about which string goes “under” at each intersection point. All the pictures we have seen so far are examples for braid diagrams by just “forgetting” about the y -axis.

There are some technical discussions to ensure that each isotopy class of geometric braids can be represented by a braid diagram. For more details, see [CK08] chapter 1.2.2.

Lastly, we define the *elementary (geometric) braids* σ_i^+ and σ_i^- to be those braids which are represented by the braid diagrams in fig. 11

3.2 Through the Lens of Algebra

We now introduce the algebraic version of the braid groups, which will turn out to be isomorphic to the geometric braid groups.

Definition 3.4. The *Artin Braid Group on n strings* is

$$B_n = \langle \sigma_1, \dots, \sigma_{n-1} \mid \sigma_i \sigma_j = \sigma_j \sigma_i \text{ for all } |i - j| \geq 2, \\ \sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1} \text{ for all } i \rangle.$$

The relations in this presentation are called *braid relations*. As special cases we have that $B_1 = 1$ is the trivial group and $B_2 = \langle \sigma_1 \rangle \cong \mathbb{Z}$. For $n \geq 3$ the group is non-abelian, which will be shown later.

Note that if we have a homomorphism $f : B_n \rightarrow G$, then the elements $f(\sigma_1), \dots, f(\sigma_{n-1})$ always satisfy the braid relations. The next lemma is kind of a converse of this fact. The proof is based upon Lemma 1.2 from [CK08].

Lemma 3.5. *If $s_1, \dots, s_{n-1}$ are elements of a group G satisfying the braid relations, then there is a unique homomorphism $f : B_n \rightarrow G$, such that $s_i = f(\sigma_i)$.*

Proof. We use the universal properties of free groups and quotient groups to prove this. First consider the set of

generators $S = \{\sigma_1, \dots, \sigma_{n-1}\}$, the free group F_S and the map $g : S \rightarrow G, \sigma_i \mapsto s_i$. Then there is a unique homomorphism $\varphi : F_S \rightarrow G$ such that $\varphi \circ \text{inc} = g$, where $\text{inc} : S \hookrightarrow F_S$ is the canonical inclusion. Next, observe that for $|i - j| \geq 2$ we have

$$\varphi(\sigma_i \sigma_j) = \varphi(\sigma_i) \varphi(\sigma_j) = s_i s_j = s_j s_i = \varphi(\sigma_j \sigma_i)$$

and similarly

$$\varphi(\sigma_i \sigma_{i+1} \sigma_i) = \varphi(\sigma_{i+1} \sigma_i \sigma_{i+1}).$$

Note that B_n is a quotient of F_S , since it has the same set of generators, but with additional relations between those generators. By the universal property of quotients, this then induces a unique homomorphism f making the following diagram commute:

$$\begin{array}{ccccc} S & \xrightarrow{\text{inc}} & F_S & \twoheadrightarrow & B_n \\ & \searrow f & \downarrow \exists! \varphi & & \downarrow \exists! f \\ & & G & & \end{array}$$

□

At this point, a connection between geometric braids B_n and Artin braids B_n can be established. We first need the following result, which can be proven using isotopic braid diagrams and Reidemeister moves. For more details, see lemma 1.11 in [CK08].

Lemma 3.6. $\sigma_1^+, \dots, \sigma_{n-1}^+ \in B_n$ satisfy the braid relations.

Together with the previous lemma, this implies the existence of a unique homomorphism $\varphi_+ : B_n \rightarrow \mathcal{B}_n, \sigma_i \mapsto \sigma_i^+$. Indeed, this is even an isomorphism. A proof of this result can be found in theorem 1.12 in [CK08].

Theorem 3.7. For a choice of plus or minus there is a unique isomorphism $\varphi_{\pm} : B_n \rightarrow \mathcal{B}_n, \sigma_i \mapsto \sigma_i^{\pm}$.

Thus from now on we identify $B_n \cong \mathcal{B}_n$ via φ_+ and write $\sigma_i = \sigma_i^+$. In this notation, we have $\sigma_i^- = \sigma_i^{-1}$. We are now able to visualize algebraic relations in B_n by braid diagrams, for example see fig. 12, where the second braid relation is visualized.

As another application of lemma 3.5, we get the next result, which is exercise 1.1.1 in [CK08]. It basically tells us, that in the abelianization B_n^{ab} , all info about the braiding is lost and only the info about the amount of elementary braids needed to form the braid remains.

Proposition 3.8. There is a group homomorphism $f : B_n \rightarrow \mathbb{Z}$, such that $f(\sigma_i) = 1$ for all $i = 1, \dots, n-1$. It induces an isomorphism $\bar{f} : B_n/[B_n, B_n] \rightarrow \mathbb{Z}$.

Proof. The existence of f is given by applying lemma 3.5 to $s_i = 1, i = 1, \dots, n-1$. By the universal property of

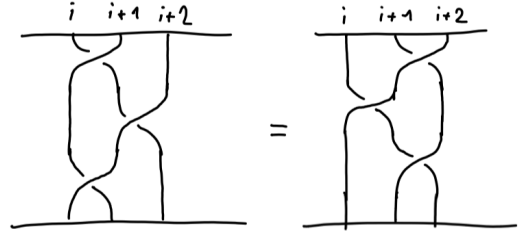


Figure 12: The second braid relation $\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$.

the abelianization, there is a unique homomorphism $\bar{f}$ making the following diagram commute:

$$\begin{array}{ccc} B_n & \xrightarrow{f} & \mathbb{Z} \\ p \downarrow & \nearrow \exists! \bar{f} & \\ B_n^{\text{ab}} & & \end{array}$$

where $B_n^{\text{ab}} = B_n/[B_n, B_n]$. Surjectivity of f is clear, since 1 generates $\mathbb{Z}$ and $f(p(\sigma_i)) = f(\sigma_i) = 1$. For injectivity, let $\bar{f}(p(a)) = \bar{f}(p(b))$ for some $a, b \in B_n$. Then $f(a) = f(b) =: k$, i.e. a and b consist of the same “amount” of elementary braids:

$$a = \prod_{j=1}^k \sigma_{i_j}, \quad b = \prod_{j=1}^k \sigma_{n_j}$$

for some $i_j, n_j \in \{1, \dots, n-1\}$. Now consider the images $\bar{\sigma}_i := p(\sigma_i)$ of elementary braids in B_n^{ab} . Using commutativity and the braid relations we obtain

$$\bar{\sigma}_i^2 \bar{\sigma}_{i+1} = \bar{\sigma}_i \bar{\sigma}_{i+1} \bar{\sigma}_i = \bar{\sigma}_{i+1} \bar{\sigma}_i \bar{\sigma}_{i+1} = \bar{\sigma}_{i+1}^2 \bar{\sigma}_i,$$

which is equivalent to

$$\bar{\sigma}_i \bar{\sigma}_{i+1}^{-1} = \bar{\sigma}_i^2 \bar{\sigma}_{i+1} \bar{\sigma}_{i+1}^{-2} \bar{\sigma}_i^{-1} = 1.$$

But this implies $p(\sigma_i) = \bar{\sigma}_i = \bar{\sigma}_{i+1} = p(\sigma_{i+1})$ and since this holds for all $i = 1, \dots, n-1$, we obtain $p(a) = p(b)$, finishing the proof. □

We will use this result again at the very end of this document to deduce the main result, corollary 4.7.

3.3 Connections to Symmetric Groups

After definition 3.1, we talked about the underlying permutation of a braid, which is the permutation one gets when following each individual string from start to end. For example, the braid in fig. 5 has the underlying permutation $(3, 4, 1, 2)$. This suggests that there is a homomorphism from B_n to the group of permutations of n

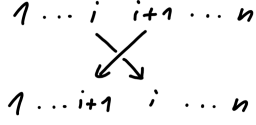


Figure 13: The simple transposition s_i .

$$\begin{pmatrix} s_i & \begin{array}{c} 1 \dots i \quad i+1 \dots j \quad j+1 \dots n \\ \swarrow \quad \searrow \\ 1 \dots i+1 \quad i \dots j \quad j+1 \dots n \end{array} \\ s_j & \begin{array}{c} 1 \dots i+1 \quad i \dots j+1 \quad j \dots n \\ \swarrow \quad \searrow \\ 1 \dots i+1 \quad i \dots j+1 \quad j \dots n \end{array} \end{pmatrix} = \begin{pmatrix} s_j & \begin{array}{c} 1 \dots i \quad i+1 \dots j+1 \quad j \dots n \\ \swarrow \quad \searrow \\ 1 \dots i+1 \quad i \dots j+1 \quad j \dots n \end{array} \\ s_i & \begin{array}{c} 1 \dots i+1 \quad i \dots j+1 \quad j \dots n \\ \swarrow \quad \searrow \\ 1 \dots i+1 \quad i \dots j+1 \quad j \dots n \end{array} \end{pmatrix}$$

$$\begin{pmatrix} s_i & \begin{array}{c} 1 \dots i \quad i+1 \quad i+2 \dots n \\ \swarrow \quad \searrow \\ i+1 \quad i \quad i+2 \end{array} \\ s_{i+1} & \begin{array}{c} i+1 \quad i \quad i+2 \\ \swarrow \quad \searrow \\ i+1 \quad i+2 \quad i \end{array} \\ s_i & \begin{array}{c} i+1 \quad i+2 \quad i \\ \swarrow \quad \searrow \\ 1 \dots i+2 \quad i+1 \quad i \dots n \end{array} \end{pmatrix} = \begin{pmatrix} s_{i+1} & \begin{array}{c} 1 \dots i \quad i+1 \quad i+2 \dots n \\ \swarrow \quad \searrow \\ i+1 \quad i+2 \quad i+1 \end{array} \\ s_i & \begin{array}{c} i+1 \quad i+2 \quad i+1 \\ \swarrow \quad \searrow \\ i+2 \quad i \quad i+1 \end{array} \\ s_{i+1} & \begin{array}{c} i+2 \quad i+1 \quad i \\ \swarrow \quad \searrow \\ 1 \dots i+2 \quad i+1 \quad i \dots n \end{array} \end{pmatrix}$$

Figure 14: $s_i s_j = s_j s_i$ for $|i - j| \geq 2$ and $s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}$.

elements Σ_n . The latter is generated by the *simple transpositions* $s_1, \dots, s_{n-1}$, where s_i permutes i and $i+1$, see fig. 13. Since the simple transpositions $s_1, \dots, s_{n-1}$ satisfy the braid relations, as depicted in fig. 14, and also generate Σ_n , there is a unique surjective group homomorphism $\pi : B_n \rightarrow \Sigma_n, \pi(\sigma_i) = s_i$. Injectivity fails, since $\pi(\sigma_i^2) = s_i^2 = 1$ but $\sigma_i^2 \neq 1$.

This homomorphism can be used to show that B_n is non-abelian for $n \geq 3$. The proof is taken from [CK08], lemma 1.3.

Proposition 3.9. *The group B_n is non-abelian for $n \geq 3$.*

Proof. Looking at Σ_n for $n \geq 3$, we have $s_1 s_2 = (2, 3, 1) \neq (3, 1, 2) = s_2 s_1$. By construction of π , we then know that $\sigma_1 \sigma_2 \neq \sigma_2 \sigma_1$, since otherwise we would have

$$s_1 s_2 = \pi(\sigma_1 \sigma_2) = \pi(\sigma_2 \sigma_1) = s_2 s_1.$$

□

Furthermore, the projection π allows us to define the notion of *pure braids*, which will be the main topic for the upcoming chapters.

Definition 3.10. The *Pure Braid Group* on n strands is

$$P_n := \ker(\pi : B_n \rightarrow \Sigma_n).$$

These correspond to those geometric braids, where a string starting at $(i, 0, 0)$ has its endpoint at $(i, 0, 1)$. For example, the braids depicted in fig. 18 are pure.

3.4 Through the Lens of Algebraic Topology

Using the pure braids introduced in the previous chapter, we can now establish a connection between geometric braids and the fundamental group of a specific topological space.

Definition 3.11. The *configuration space of n points in $\mathbb{R}^2$* is

$$\mathcal{F}_n(\mathbb{R}^2) := \{(x_1, \dots, x_n) \in (\mathbb{R}^2)^n \mid x_i \neq x_j \text{ for } i \neq j\}.$$

It inherits the structure of a topological space as a subspace of $(\mathbb{R}^2)^n$.

The following statements are based upon [CK08] chapter 1.4.1. Consider $\mathcal{F}_n(\mathbb{R}^2)$ with $q_n := ((1, 0), (2, 0), \dots, (n, 0))$ as its basepoint. Given a pure geometric braid b let $u_i(t)$ be the unique point in $\mathbb{R}^2$ such that the i -th string of b meets $\mathbb{R}^2 \times t$ at $(u_i(t), t)$. Furthermore, denote as $\Omega(\mathcal{F}_n(\mathbb{R}^2))$ the set of all loops in $\mathcal{F}_n(\mathbb{R}^2)$ with respect to the basepoint q_n and as $\mathcal{B}_n^{\text{pure}}$ the set of pure geometric braids, i.e. those braids where the underlying permutation is trivial. With this notation, consider

$$\begin{aligned} \varphi : \mathcal{B}_n^{\text{pure}} &\rightarrow \Omega(\mathcal{F}_n(\mathbb{R}^2)), \\ b &\mapsto (t \mapsto (u_1(t), \dots, u_n(t))). \end{aligned}$$

Its inverse is given by

$$\begin{aligned} \varphi^{-1} : \Omega(\mathcal{F}_n(\mathbb{R}^2)) &\rightarrow \mathcal{B}_n^{\text{pure}}, \\ (\alpha_1, \dots, \alpha_n) &\mapsto \bigcup_{i=1}^n \bigcup_{t \in I} (\alpha_i(t), t). \end{aligned}$$

Intuitively, φ yields a loop in $\mathcal{F}_n(\mathbb{R}^2)$ by following along the individual geometric strings for each time t . On the other hand, φ^{-1} collects the images of the individual paths in $\mathcal{F}_n(\mathbb{R}^2)$ and arranges them in a manner that is similar to the “stacking” of $\mathbb{R}^2$ -layers discussed in the introduction.

This yields a bijective correspondence $\mathcal{B}_n^{\text{pure}} \cong \Omega(\mathcal{F}_n(\mathbb{R}^2))$. Without going into further detail, one can show that under this bijection, isotopies of braids correspond to homotopies of loops. Furthermore, the group structure, which is multiplication of braids on one hand and concatenation of loops on the other hand, is also respected. Thus we get the following result.

Proposition 3.12. *We have $P_n \cong \pi_1(\mathcal{F}_n(\mathbb{R}^2))$.*

Now what about non-pure braids B_n ? It turns out, that they correspond to the fundamental group of the

unlabeled configuration space

$$\mathcal{C}_n(\mathbb{R}^2) = \mathcal{F}_n(\mathbb{R}^2)/\Sigma_n$$

where Σ_n acts on $\mathcal{F}_n(\mathbb{R}^2)$ by permuting the coordinates. Intuitively, a loop in this space is allowed to go from one configuration to another configuration as long as the set of start-points is the same as the set of end-points. Considering the fundamental group $\pi_1(\mathcal{C}_n(\mathbb{R}^2))$, this translates to not only including pure braids but also non-pure ones, thus

$$\pi_1(\mathcal{C}_n(\mathbb{R}^2)) \cong B_n.$$

For more details, see chapter 1.4.3 in [CK08].

Before continuing further, let us establish another kind of configuration space and some additional facts about these spaces. First we generalize the definition of $\mathcal{F}_n(\mathbb{R}^2)$ by specifying a set of “prohibited” points in $\mathbb{R}^2$.

Definition 3.13. Fix a set $Q_m = \{p_1, \dots, p_m\} \in \mathbb{R}^2$ of m points, then define

$$\mathcal{F}_{m,n}(\mathbb{R}^2) = \mathcal{F}_n(\mathbb{R}^2 \setminus Q_m).$$

Topologically (at least with respect to homotopy groups), the exact choice of the m points in Q_m does not matter (as long as they are pairwise different). Also note, that

$$\mathcal{F}_{0,n}(\mathbb{R}^2) = \mathcal{F}_n(\mathbb{R}^2), \quad \mathcal{F}_{m,1}(\mathbb{R}^2) = \mathbb{R}^2 \setminus Q_m.$$

These configuration spaces with prohibited points will prove useful when analyzing the structure of pure braid groups. This is due to the next two statements, proofs of which can be found in [CK08] lemmas 1.26 and 1.27.

Lemma 3.14. For $n > r \geq 1$, the map

$$\begin{aligned} p : \mathcal{F}_n(\mathbb{R}^2) &\rightarrow \mathcal{F}_r(\mathbb{R}^2), \\ (u_1, \dots, u_n) &\mapsto (u_1, \dots, u_r) \end{aligned}$$

is a locally trivial fibration with fiber $\mathcal{F}_{r,n-r}(\mathbb{R}^2)$.

Applied to $\mathbb{R}^2 \setminus Q_m$ this yields:

Lemma 3.15. For $m \geq 0, n > r \geq 1$, the map

$$\begin{aligned} p : \mathcal{F}_{m,n}(\mathbb{R}^2) &\rightarrow \mathcal{F}_{m,r}(\mathbb{R}^2), \\ (u_1, \dots, u_n) &\mapsto (u_1, \dots, u_r) \end{aligned}$$

is a locally trivial fibration with fiber $\mathcal{F}_{m+r,n-r}(\mathbb{R}^2)$.

The proof for the next result is taken from [CK08], lemma 1.28, and it is stated here, mainly because it illustrates the use of the homotopy sequence induced by the fibration considered above.

Lemma 3.16. For any $m \geq 0, n \geq 1$, we have

$$\pi_i(\mathcal{F}_{n,m}(\mathbb{R}^2)) = 1$$

for all $i \geq 2$.

Proof. Consider the fibration

$$\mathcal{F}_{m+1,n-1}(\mathbb{R}^2) \hookrightarrow \mathcal{F}_{m,n}(\mathbb{R}^2) \twoheadrightarrow \mathcal{F}_{m,1}(\mathbb{R}^2) = \mathbb{R}^2 \setminus Q_m.$$

Note that

$$\mathbb{R}^2 \setminus Q_m \simeq \bigvee_{i=1}^m \mathbb{S}^1.$$

Since its universal cover is a tree and thus contractible, we have

$$\pi_i(\mathcal{F}_{m,1}(\mathbb{R}^2)) \cong \pi_i(\mathbb{R}^2 \setminus Q_m) \cong 1$$

for all $i \geq 2$.

The fibration above induces the homotopy sequence

$$\begin{array}{ccccccc} & & & & \cdots & \longrightarrow & \underbrace{\pi_{i+1}(\mathcal{F}_{m,1}(\mathbb{R}^2))}_{\cong 1} \\ & & & & & \swarrow & \\ \pi_i(\mathcal{F}_{m+1,n-1}(\mathbb{R}^2)) & \xrightarrow{\cong} & \pi_i(\mathcal{F}_{m,n}(\mathbb{R}^2)) & \longrightarrow & \underbrace{\pi_i(\mathcal{F}_{m,1}(\mathbb{R}^2))}_{\cong 1} \\ & & & & \nwarrow & & \\ \cdots & & & & & & \end{array}$$

where $i \geq 2$ in this diagram. Inductively, this leads to

$$\begin{aligned} \pi_i(\mathcal{F}_{m,n}(\mathbb{R}^2)) &\cong \pi_i(\mathcal{F}_{m+n-1,1}(\mathbb{R}^2)) \\ &\cong \pi_i(\mathbb{R}^2 \setminus Q_{m+n-1}) \\ &\cong 1 \end{aligned}$$

for all $i \geq 2$, finishing the proof. $\square$

Before we leave this topic for now, let us have a look at the special case $r = n - 1$. The following discussion is based upon the proof of theorem 1.16 from [CK08]. Lemma 3.14 tells us, that

$$\mathcal{F}_{n-1,1}(\mathbb{R}^2) \xhookrightarrow{\iota} \mathcal{F}_n(\mathbb{R}^2) \twoheadrightarrow \mathcal{F}_{n-1}(\mathbb{R}^2)$$

is a locally trivial fibration, where p maps $(u_1, \dots, u_n)$ to $(u_1, \dots, u_{n-1})$, so we just “forget” about the last point. For a given point $(u_1, \dots, u_{n-1}) \in \mathcal{F}_{n-1}(\mathbb{R}^2)$, the fiber is

$$\begin{aligned} p^{-1}(\{(u_1, \dots, u_{n-1})\}) \\ = \{(u_1, \dots, u_{n-1}, a) \mid a \in \mathbb{R}^2, a \neq u_i \ \forall i = 1, \dots, n-1\}, \end{aligned}$$

which is the same as

$$\mathcal{F}_{n-1,1}(\mathbb{R}^2) = \mathcal{F}_1(\mathbb{R}^2 \setminus Q_{n-1}) = \mathbb{R}^2 \setminus Q_{n-1}.$$

if we choose $Q_{n-1} = \{u_1, \dots, u_{n-1}\}$. So, visually speaking, after choosing a point $a \in \mathbb{R}^2 \setminus Q_{n-1}$, the map ι “fills in” the missing points $u_1, \dots, u_{n-1}$, see fig. 15.

Concluding this section, let us have a look at the fundamental groups of the three spaces occurring in the fibration. Recalling the visualization technique from the

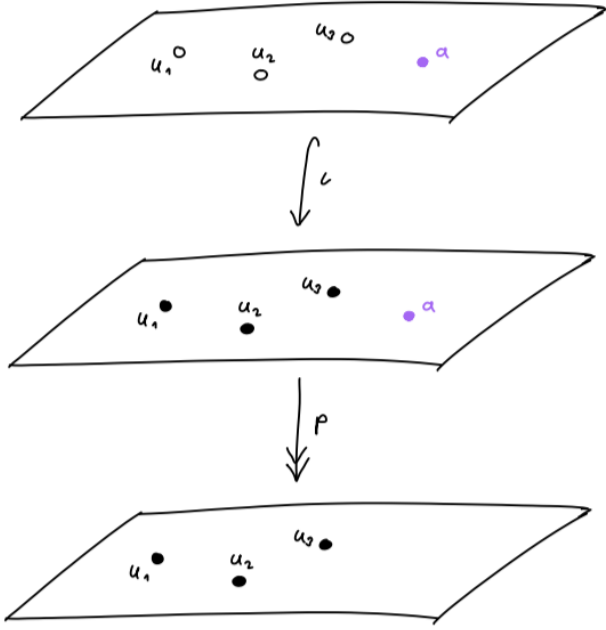


Figure 15: Visualization of the fibration from lemma 3.14 for $n = 4, r = n - 1 = 3$.

introductory example, loops in the spaces from the fibration sequence can be drawn as in fig. 16. Regarding a loop in $\pi_1(\mathcal{F}_{n-1,1}(\mathbb{R}^2))$, this can be understood as loop in $\mathbb{R}^2$ starting and ending at the basepoint a , but avoiding the set $\{u_1, \dots, u_{n-1}\}$. The induced map $\iota_{\#}$ then maps this loop to a pure braid in $\pi_1(\mathcal{F}_n(\mathbb{R}^2))$ where only the string connecting a to itself does something non-trivial. Finally, the induced map $p_{\#}$ then corresponds to deleting the string connecting a to itself and keeps only the strings connecting the individual points u_i . It is helpful to keep this pictures in mind for the following two sections.

4 Structural Properties of Braid Groups

After establishing different versions of (pure) braid groups and seeing that they are all isomorphic, we can move to the main part of this document. In this chapter, we will see various structural results about the pure braid group P_n , which eventually can be used to show the main result, which states $B_n \not\cong B_m$ for $n \neq m$. The definitions and results in this chapter are based chapter 1.3 in [CK08].

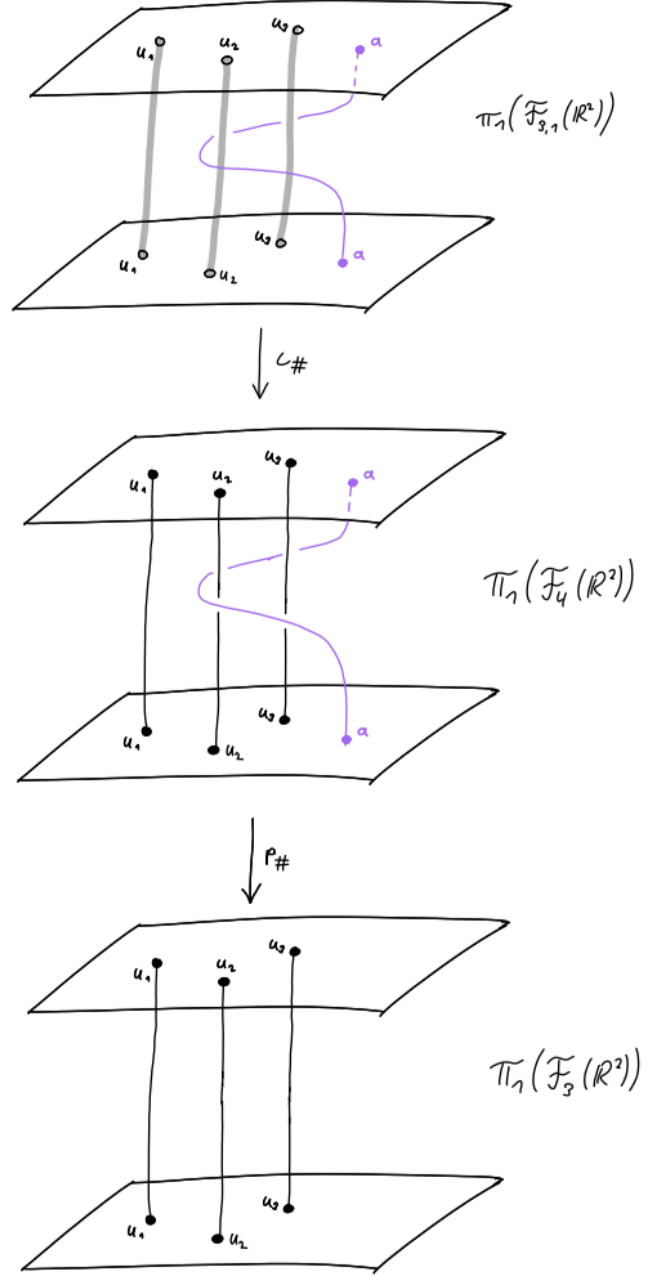


Figure 16: Visualization of the maps on fundamental groups induced by the fibration sequence.

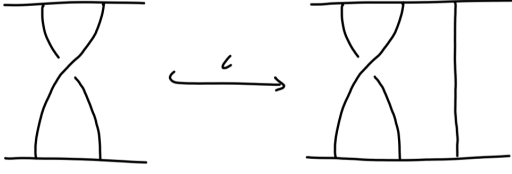


Figure 17: The inclusion $B_2 \hookrightarrow B_3$.

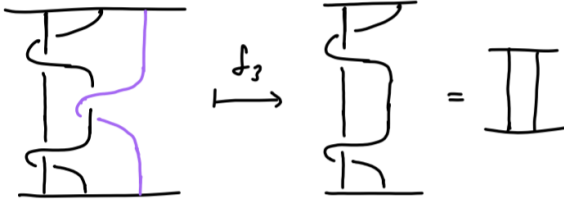


Figure 18: An element of U_3 .

4.1 Inclusions, Forgetting Homomorphisms and Normal Form

There is a natural inclusion $\iota : B_n \rightarrow B_{n+1}, \sigma_i \mapsto \sigma_i$. Geometrically, this inclusion can be understood as just adding a string on the right that is unlinked to the existing strings, see fig. 17. For pure braids there is also a forgetting homomorphism $f_n : P_n \rightarrow P_{n-1}$ which just deletes the n -th string. Since the natural inclusion restricts to $\iota : P_n \hookrightarrow P_{n+1}$, we have $f_n \circ \iota = \text{id}_{P_{n-1}}$. This implies injectivity of ι (restricted to P_n) and surjectivity of f_n . Furthermore, using braid diagrams and isotopies, it can be shown that $\iota : B_n \rightarrow B_{n+1}$ is injective. For details, see corollary 1.14 in [CK08].

Thus B_n can be viewed as a subgroup of B_{n+1} via ι . Note that we have the following commutative diagram:

$$\begin{array}{ccc} B_n & \xrightarrow{\pi} & \Sigma_n \\ \downarrow \iota & & \downarrow \\ B_{n+1} & \xrightarrow{\pi} & \Sigma_{n+1} \end{array}$$

If we denote

$$U_n := \ker f_n,$$

then U_n consists of exactly those pure braids in P_n where removing the n -th string “disentangles the braid”, see fig. 18. We then obtain the split exact sequence

$$1 \longrightarrow U_n \hookrightarrow P_n \xrightarrow{f_n} P_{n-1} \longrightarrow 1.$$

$\quad \quad \quad \swarrow \iota$

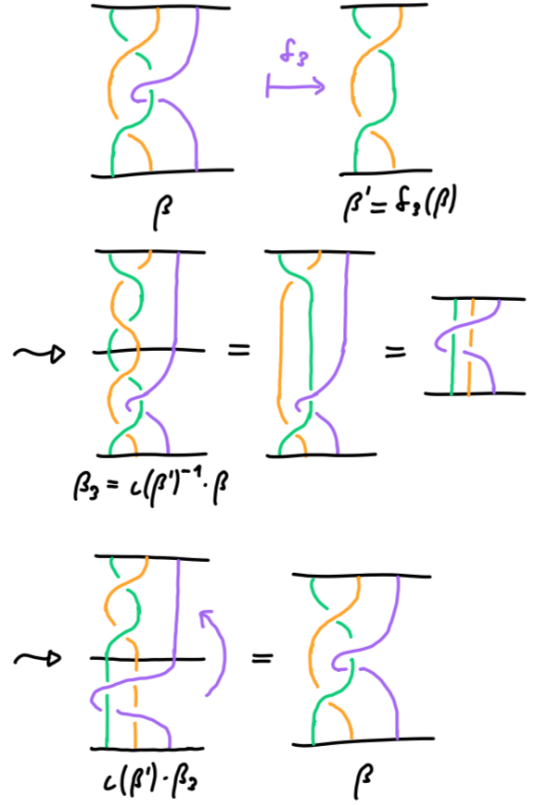


Figure 19: The equality $\beta = \iota(\beta')\beta_3$ for a specific braid $\beta \in P_3$.

Thus we have that P_n is the semidirect product of U_n and P_{n-1} , i.e.

$$P_n = U_n \rtimes P_{n-1}.$$

From this it follows that a pure braid $\beta \in P_n$ has a unique expansion $\beta = \iota(\beta')\beta_n$ with $\beta' \in P_{n-1}, \beta_n \in U_n$, namely $\beta' = f_n(\beta)$ and $\beta_n = \iota(\beta')^{-1}\beta$, see fig. 19 for an example. Inductively, β can be written as

$$\beta = \beta_2\beta_3 \cdots \beta_n,$$

where $\beta_j \in U_j \subseteq P_j \subseteq P_n$.

This is the *combed* or *normal form* of β . As discussed in [Abr17] theorems 18.2 and 18.3, this implies that the word problem is solvable for braid groups: If two braids are equivalent, they have the same normal form. So in order to check if a braid is trivial, one can calculate its normal form and see if this form is the identity. If not, the braid is not trivial.

Additionally to being useful for solving the word problem, the fact that $P_n = U_n \rtimes P_{n-1}$ will be used again in section 4.4 when analyzing the center of P_n .

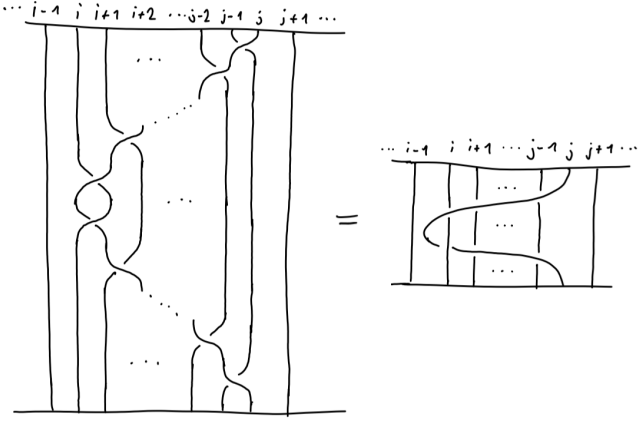


Figure 20: The pure braid $A_{i,j}$.

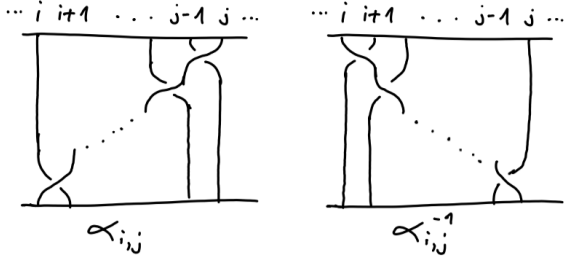


Figure 21: The non-pure braid $\alpha_{i,j}$ and its inverse.

4.2 Elementary Pure Braids

We now introduce certain pure braids in P_n that will turn out to be the generators of P_n . In order to show this, one can use arguments from algebraic topology using the results about fibrations which we established at the end of chapter 3.4. This is interesting, because often topologists use a lot of algebra to solve topological problems, but this time topological tools are used to deduct algebraic results.

For $1 \leq i < j \leq n$ consider the pure n -string braid

$$A_{i,j} := \sigma_{j-1}\sigma_{j-2} \cdots \sigma_{i+1}\sigma_i^2\sigma_{i+1}^{-1} \cdots \sigma_{j-2}^{-1}\sigma_{j-1}^{-1},$$

see fig. 20. These elemental pure braids $A_{i,j}$ are conjugate to each other in the group of (not necessarily pure) braids B_n . To see this, set

$$\alpha_{i,j} := \sigma_{j-1}\sigma_{j-2} \cdots \sigma_i$$

for $1 \leq i < j \leq n$, see fig. 21. Then for $1 \leq i < j < k \leq n$

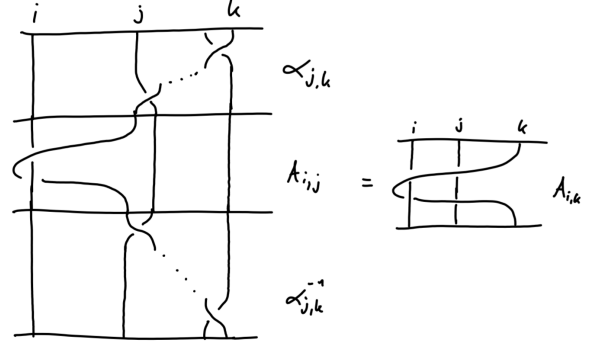


Figure 22: Conjugation of $A_{i,j}$ yields $A_{i,k}$.

we have

$$\alpha_{j,k}A_{i,j}\alpha_{j,k}^{-1} = A_{i,k} \quad \text{and} \quad \alpha_{i,k}A_{i,j}\alpha_{i,k}^{-1} = A_{j,k},$$

see fig. 22.

We will now see that the elements $\{A_{i,n}\}_{i=1,\dots,n-1}$ generate the subgroup $U_n \leq P_n$ which was introduced in the last chapter. The proof uses results about the fibration $\mathcal{F}_n(\mathbb{R}^2) \rightarrow \mathcal{F}_{n-1}(\mathbb{R}^2)$ discussed in chapter 3.4 and is taken from [CK08], theorem 1.16.

Theorem 4.1. *For $n \geq 2$, the group U_n is free on the $n-1$ generators $\{A_{i,n}\}_{i=1,\dots,n-1}$, i.e.*

$$U_n = \langle A_{i,n} \mid i \in \{1, \dots, n-1\} \rangle.$$

Proof. By lemma 3.14, we have the locally trivial fibration

$$\mathcal{F}_{n-1,1}(\mathbb{R}^2) \xhookrightarrow{\iota} \mathcal{F}_n(\mathbb{R}^2) \xrightarrow{p} \mathcal{F}_{n-1}(\mathbb{R}^2),$$

where p maps $(u_1, \dots, u_n) \mapsto (u_1, \dots, u_{n-1})$.

Note that $\pi_0(\mathcal{F}_{n-1,1}(\mathbb{R}^2)) \cong \pi_0(\mathbb{R}^2 \setminus Q_{n-1}) \cong 1$, since π_0 is represented by path-connected components, and $\pi_2(\mathcal{F}_{n-1,1}(\mathbb{R}^2)) \cong 1$ by lemma 3.16. Thus the homotopy sequence of the aforementioned fibration yields the short exact sequence

$$\begin{array}{c} 1 \\ \downarrow \\ \pi_1(\mathcal{F}_{n-1,1}(\mathbb{R}^2)) \\ \downarrow \\ \pi_1(\mathcal{F}_n(\mathbb{R}^2)) \\ \downarrow \\ \pi_1(\mathcal{F}_{n-1}(\mathbb{R}^2)) \\ \downarrow \\ 1 \end{array}$$

Recall from chapter 3.4 that there is a bijection $\varphi : P_n \xrightarrow{\cong} \pi_1(\mathcal{F}_n(\mathbb{R}^2))$ that respects the group structure. In this chapter we also discussed what the induced maps from the fibration sequence do for elements of the fundamental group. Under this identification, the induced map $p_\#$ corresponds to the forgetting homomorphism f_n . Thus we have the following commutative diagram

$$\begin{array}{ccc}
1 & & 1 \\
\downarrow & & \downarrow \\
\pi_1(\mathcal{F}_{n-1,1}(\mathbb{R}^2)) & & U_n \\
\downarrow \iota_\# & & \downarrow \\
\pi_1(\mathcal{F}_n(\mathbb{R}^2)) & \xrightarrow[\cong]{\varphi^{-1}} & P_n \\
\downarrow p_\# & & \downarrow f_n \\
\pi_1(\mathcal{F}_{n-1}(\mathbb{R}^2)) & \xrightarrow[\cong]{\varphi^{-1}} & P_{n-1} \\
\downarrow & & \downarrow \\
1 & & 1
\end{array}$$

with exact columns. From this we can deduce

$$U_n = \ker f_n = \text{im}(\varphi^{-1} \circ \iota_\#),$$

so it all boils down to understanding what the induced map $\iota_\#$ does.

We can choose $Q_{n-1} \in \mathbb{R}^2$ as $\{(1,0), (2,0), \dots, (n-1,0)\}$ and choose $a_0 = (n,0)$ as the basepoint of $\mathbb{R}^2 \setminus Q_{n-1}$. Since $\mathbb{R}^2 \setminus Q_{n-1}$ deformation retracts to $\bigvee_{i=1}^{n-1} \mathbb{S}^1$, the group $\pi_1(\mathcal{F}_{n-1,1}(\mathbb{R}^2)) = \pi_1(\mathbb{R}^2 \setminus Q_{n-1})$ is free on $n-1$ generators $x_1, \dots, x_{n-1}$ as depicted in fig. 23. By definition of the initial “forgetting fibration” p , the inclusion ι is given by

$$\mathcal{F}_{n-1,1}(\mathbb{R}^2) = \mathbb{R}^2 \setminus Q_{n-1} \xrightarrow{\iota} \mathcal{F}_n(\mathbb{R}^2)$$

$$(x, y) \longmapsto ((1,0), \dots, (n-1,0), (x,y))$$

and thus the induced map $\iota_\#$ can be described as

$$\pi_1(\mathcal{F}_{n-1,1}(\mathbb{R}^2)) \xrightarrow{\iota_\#} \pi_1(\mathcal{F}_n(\mathbb{R}^2))$$

$$[\alpha] \longmapsto [\iota \circ \alpha] = [((1,0), \dots, (n-1,0), \alpha)]$$

where $[\alpha]$ denotes the homotopy class of some loop α and $((1,0), \dots, (n-1,0), \alpha)$ is the loop which is constant on its first $n-1$ entries and has the values of α in its last entry. Now let us look more closely, what this means for the generators x_i : We have

$$\iota_\#(x_i) = [((1,0), \dots, (n-1,0), x_i)],$$

which under the map φ^{-1} becomes $\varphi^{-1} \circ \iota_\#(x_i) = A_{i,n}$, see fig. 24. Using the fact that the x_i generate $\pi_1(\mathcal{F}_{n-1,1}(\mathbb{R}^2))$ and that $\iota_\#$ is injective by exactness, we

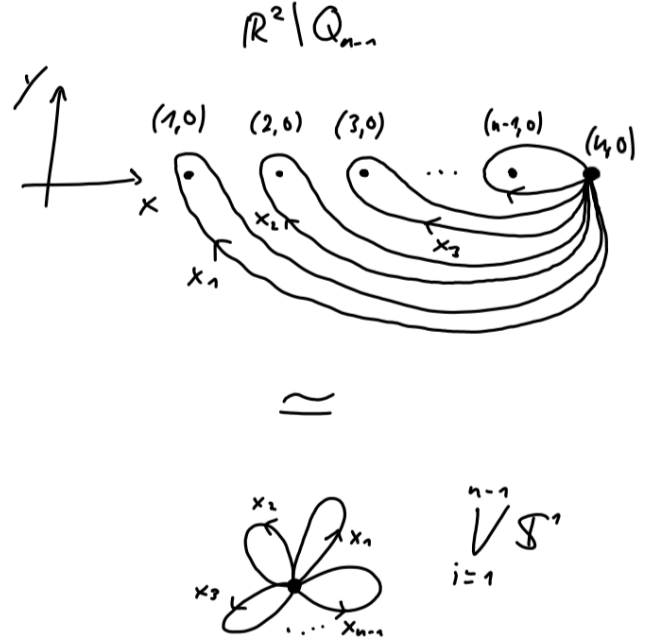


Figure 23: The generators of $\pi_1(\mathcal{F}_{n-1,1}(\mathbb{R}^2))$.

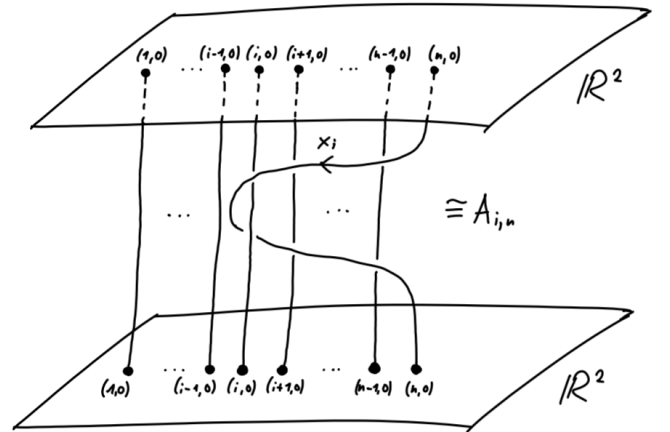


Figure 24: The image $\varphi^{-1} \circ \iota_\#(x_i) = A_{i,n}$.

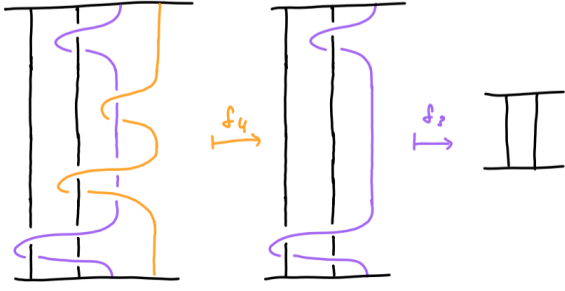


Figure 25: A pure braid in $U_4^{(2)}$.

obtain the desired statement

$$U_n = \text{im}(\varphi^{-1} \circ \iota_{\#}) = \langle A_{i,n} \rangle_{i=1, \dots, n-1}.$$

□

Note that in particular this means that U_n is torsion-free, which is relevant for the proof at the end of the next chapter.

4.3 More Structural Results about Pure Braids

In this chapter, the goal is to prove that P_n is torsion-free. This fact will later be used for finding out what the center of P_n is. In order to prove torsion-freeness, we first show that P_n admits a filtration of specific normal subgroups, which we will now define.

First, consider concatenations of i forgetting homomorphisms

$$P_n \xrightarrow{f_n} P_{n-1} \xrightarrow{f_{n-1}} \dots \xrightarrow{f_{n-i+2}} P_{n-i+1} \xrightarrow{f_{n-i+1}} P_{n-i}.$$

Define the subgroups

$$U_n^{(i)} := \ker(f_{n-i+1} \circ f_{n-i+2} \circ \dots \circ f_{n-1} \circ f_n)$$

where we set $U_n(0) := 1$. Note that $U_n^{(1)} = U_n$ which we have studied before. Intuitively, applying $f_{n-i+1} \circ \dots \circ f_n$ we “forget” what happens in the last i strings of pure braids on n strings and thus $U_n^{(i)}$ contains exactly those strings, where removing the last i strings “disentangles the braid”, see fig. 25 for an element in $U_4^{(2)}$. Note that each $U_n^{(i)}$ is a normal subgroup as a kernel of some homomorphism. Furthermore we have $U_n^{(i-1)} \subseteq U_n^{(i)}$, so we can consider quotients $U_n^{(i)} / U_n^{(i-1)}$. What does an element of the quotient look like? Two braids $\beta, \beta' \in U_n^{(i)}$ represent the same braid in $U_n^{(i)} / U_n^{(i-1)}$ if there is a braid $\gamma \in U_n^{(i-1)}$ such that $\beta = \beta' \gamma$. For example, $A_{2,3} A_{3,4}$ and $A_{2,3}$ represent the same braid in $U_4^{(2)} / U_4^{(1)}$, see fig. 26.

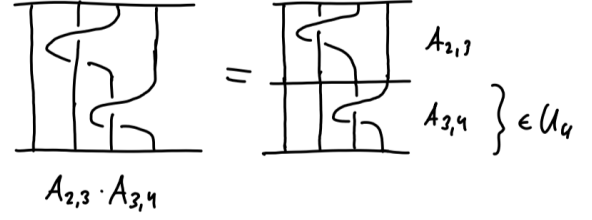


Figure 26: $A_{2,3} A_{3,4}$ and $A_{2,3}$ are equivalent in $U_4^{(2)} / U_4^{(1)}$.

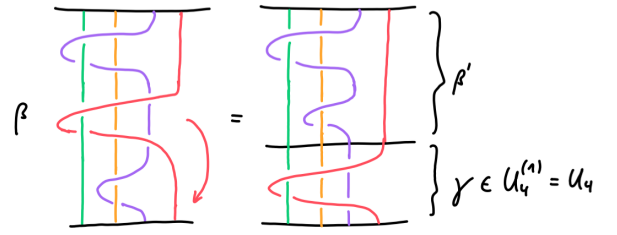


Figure 27: β and β' only differing by an element in $U_n^{(1)}$.

For another example where two braids $\beta, \beta' \in U_n^{(2)} / U_n^{(1)}$ are equivalent for the case $n = 4$, see fig. 27. With this in mind, for elements in the quotient it doesn't matter what happens in the last $(i-1)$ strings, so we could view elements in the quotient as represented by braids on $n-i+1$ strings, implying $U_n^{(i)} / U_n^{(i-1)} \cong U_{n-i+1}$. This is indeed the case: By definition of $U_n^{(i-1)}$ we have

$$\underbrace{f_{n-i+2} \circ \dots \circ f_n}_{=: F}(U_n^{(i)}) \subseteq \ker f_{n-i+1} = U_{n-i+1}.$$

By the universal property of the quotient there then exists a unique $\bar{F}$ making the following diagram commute

$$\begin{array}{ccc} U_n^{(i)} & \xrightarrow{F} & U_{n-i+1} \\ \downarrow & \nearrow \bar{F} & \\ U_n^{(i)} / U_n^{(i-1)} & & \end{array}$$

and since $U_n^{(i-1)} = \ker F$, the homomorphism $\bar{F}$ is injective. From an earlier discussion about the fact that $P_n = U_n \rtimes P_{n-1}$ we know that the inclusion $\iota_n : P_{n-1} \hookrightarrow P_n$ is a section of f_n , so if we take some $\beta \in U_{n-i+1} \subseteq P_{n-i+1}$, we first see that

$$\begin{aligned} f_{n-i+1} \circ f_{n-i+2} \circ \dots \circ f_n \circ \iota_n \circ \dots \circ \iota_{n-i+2}(\beta) \\ = f_{n-i+1}(\beta) = 1, \end{aligned}$$

meaning $\iota_n \circ \dots \iota_{n-i+2}(\beta) \in U_n^{(i)}$ and also

$$F \circ \iota_n \circ \dots \iota_{n-i+2}(\beta) = \beta,$$

implying surjectivity of F , which also means that $\bar{F}$ is surjective, making it an isomorphism.

Those results leads to the following corollary of theorem 4.1, which is corollary 1.17 in [CK08].

Corollary 4.2. *The pure braid group P_n admits a normal filtration*

$$1 = U_n^{(0)} \subseteq U_n^{(1)} \subseteq \dots \subseteq U_n^{(n-1)} = P_n$$

such that $U_n^{(i)} / U_n^{(i-1)} \cong U_{n-i+1}$ is a free group on $n-i$ generators for all i .

From this follows directly the next result, the proof of which is taken from corollary 1.18 in [CK08]. The idea is to show inductively, using the normal filtration from the previous corollary, that a torsion-element β must lie in $U_n^{(n-2)}$, then it must also lie in $U_n^{(n-3)}$ and so on, until we arrive at $U_n^{(1)} = U_n$, which is torsion-free.

Corollary 4.3. *P_n is torsion-free.*

Proof. Assume that β is a torsion-element, i.e. there is a positive integer k such that $\beta^k = 1$. Then $\beta^k = 1$ in $P_n / U_n^{(n-2)} = U_n^{(n-1)} / U_n^{(n-2)} \cong U_2$. Since the latter group is free and thus torsion-free, the image of β in this quotient must be equal to 1, which means that β is an element in $U_n^{(n-2)}$.

But likewise $\beta^k = 1$ in $U_n^{(n-2)} / U_n^{(n-3)}$, so β is an element in $U_n^{(n-3)}$. Continuing inductively, we obtain that β is an element of $U_n^{(1)} = U_n$ which is torsion-free as a free group and thus $\beta = 1$. $\square$

Another consequence of theorem 4.1 uses the normal form of pure braids: Recall that each pure braid $\beta \in P_n$ can be written as

$$\beta = \beta_2 \beta_3 \dots \beta_n,$$

where $\beta_j \in U_j \subseteq P_j \subseteq P_n$. Combining this with the fact that U_j is free on the generators $A_{i,j}$ with $i = 1, \dots, j-1$, we directly obtain the following result, which is stated as corollary 1.19 in [CK08].

Corollary 4.4. *P_n is generated by the $\frac{n(n-1)}{2}$ elements $A_{i,j}$ with $1 \leq i < j \leq n$.*

In appendix 1 of [Han89], it is shown that the defining

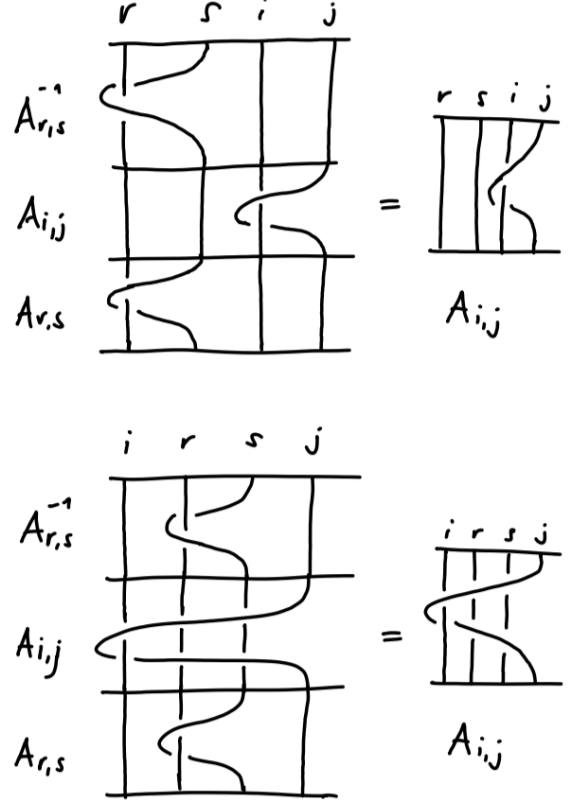


Figure 28: The relation $A_{r,s}^{-1} A_{i,j} A_{r,s} = A_{i,j}$ for the cases $s < i$ or $i < r < s < j$.

relations for those generators are

$$A_{r,s}^{-1} A_{i,j} A_{r,s} = \begin{cases} A_{i,j}, & s < i, \\ A_{i,j}, & i < r < s < j, \\ A_{r,j} A_{i,j} A_{r,j}^{-1}, & s = i, \\ A_{r,j} A_{s,j} A_{i,j} A_{s,j}^{-1} A_{r,j}^{-1}, & i = r < s < j, \\ A_{r,j} A_{s,j} A_{r,j}^{-1} A_{s,j}^{-1} A_{i,j} A_{s,j} A_{r,j} A_{s,j}^{-1} A_{r,j}^{-1}, & r < i < s < j. \end{cases}$$

For a visualization of the first two relations, see fig 28 and for the third relation, see fig. 29.

With the help of these relations we can define a *length homomorphism* $P_n \rightarrow \mathbb{Z}^{n(n-1)/2}$ similar to the one we used in the proof of 3.8. The construction is based upon [CK08] corollary 1.20 and exercise 1.3.1.

First, define for $1 \leq i < j \leq n$ a map $\ell_{i,j}$ on the generators of P_n as

$$\ell_{i,j}(\beta) = \begin{cases} 1, & \text{if } \beta = A_{i,j}, \\ 0, & \text{if } \beta = A_{r,s}, (r,s) \neq (i,j). \end{cases}$$

Since the elements $A_{i,j}$ generate P_n this map can be extended to be a homomorphism $P_n \rightarrow \mathbb{Z}$. This homomor-

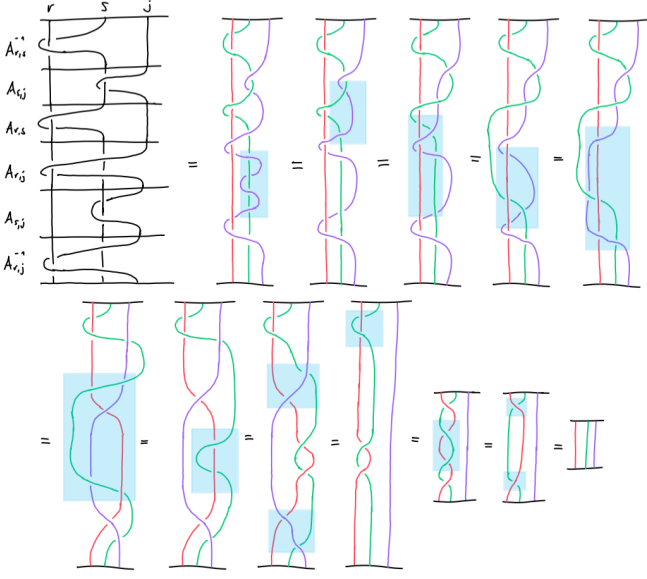


Figure 29: The relation $A_{r,s}^{-1} A_{i,j} A_{r,s} \cdot A_{r,j} A_{i,j}^{-1} A_{r,j}^{-1} = 1$ for the case $s = i$. The blue rectangles mark those spots where the changes happen.

phism is compatible with the defining relations of the generators, since in each relation for $A_{r,s}^{-1} A_{i,j} A_{r,s}$, only one $A_{i,j}$ appears and the rest is irrelevant, as they get mapped to 0. The map $\ell_{i,j}$ basically just counts, how often $A_{i,j}$ appears in a pure braid.

These individual homomorphisms can then be combined into a single homomorphism

$$L : P_n \rightarrow \mathbb{Z}^{n(n-1)/2}, \quad \beta \mapsto (\ell_{i,j}(\beta))_{1 \leq i < j \leq n}.$$

Again, similar to before when we used a length homomorphism to show $B_n^{\text{ab}} \cong \mathbb{Z}$, L can be used to figure out the abelianization of P_n . First note that this combined homomorphism is surjective: Let $(a_{i,j})_{1 \leq i < j \leq n} \in \mathbb{Z}^{n(n-1)/2}$, then define

$$\beta := A_{1,2}^{a_{1,2}} A_{1,3}^{a_{1,3}} \cdots A_{n-2,n}^{a_{n-2,n}} A_{n-1,n}^{a_{n-1,n}}.$$

Then $L(\beta) = (a_{i,j})_{1 \leq i < j \leq n}$.

Now all that is left to check is that $[P_n, P_n] \subseteq \ker L$ to get the following result.

Corollary 4.5. *The abelianization of P_n is $P_n^{\text{ab}} = P_n/[P_n, P_n] \cong \mathbb{Z}^{n(n-1)/2}$.*

Proof. Since the elements $\{A_{i,j}\}_{1 \leq i < j \leq n}$ generate P_n , they also generate P_n^{ab} . Regarding $\ker L$, first note that $[P_n, P_n] \subseteq \ker L$, since the image of L lands in an abelian group and thus the image of an element in the commutator subgroup vanishes.

For the other inclusion, let $\beta \in \ker L$, meaning $\ell_{i,j}(\beta) = 0$ for all i, j . This means, each $A_{i,j}$ appearing in the decom-

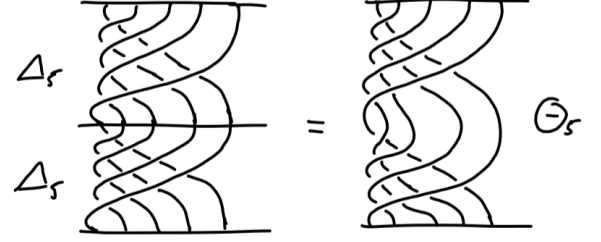


Figure 30: Half-twists and full-twist for $n = 5$.

position of β by generating elements must coincide with some $A_{i,j}^{-1}$ also appearing in β . But then the image of β under the projection $P_n \rightarrow P_n/[P_n, P_n]$ is equal to 1, implying $\beta \in [P_n, P_n]$, giving $\ker L \subseteq [P_n, P_n]$.

Recalling the universal property of abelianization, this means the map

$$\bar{L} : P_n/[P_n, P_n] \rightarrow \mathbb{Z}^{n(n-1)/2}$$

induced by L is both surjective and injective, i.e. an isomorphism. $\square$

This was not the only useful application of the length homomorphism L . We will revisit it later when analyzing the center of P_n in the next chapter.

4.4 Consequences for B_n

Finally, in this chapter the main result of the document will be discussed, namely that $B_n \not\cong B_m$ for $n \neq m$. This follows easily from the fact that the center of B_n is generated by specific pure braids called *full-twists*. Along the way, we explore various algebraic braid identities via braid diagrams. Also in this chapter we will use a lot of the results we established earlier in this document, first and foremost properties of the generators of P_n , the length homomorphisms defined in proposition 3.8 and before 4.5, torsion-freeness of P_n , as well as the projection $\pi : B_n \rightarrow \Sigma_n$.

It will turn out that the center of P_n and the center of B_n as well is generated by the *full-twist* $\theta_n \in P_n$, which consists of two *half-twists* $\Delta_n \in B_n$, i.e. $\theta_n = \Delta_n^2$, where

$$\Delta_n = (\sigma_1 \cdots \sigma_{n-1})(\sigma_1 \cdots \sigma_{n-2}) \cdots (\sigma_1 \sigma_2) \sigma_1,$$

see fig. 30. Before this is proven, however, we need to establish some equalities. First note that the image of Δ_n under the projection $B_n \rightarrow \Sigma_n$ is $(n, n-1, \dots, 1)$, so indeed $\theta_n = \Delta_n^2 \in P_n$.

One can construct θ_n from θ_{n-1} inductively by defining

$$\gamma_n := A_{1,n} A_{2,n} \cdots A_{n-1,n} \in P_n,$$

see fig. 31. Then $\theta_n = \iota(\theta_{n-1})\gamma_n$, which is visualized

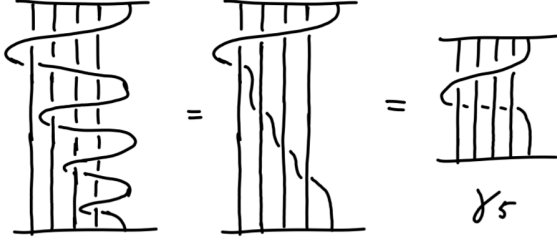


Figure 31: Visualization of γ_5 .

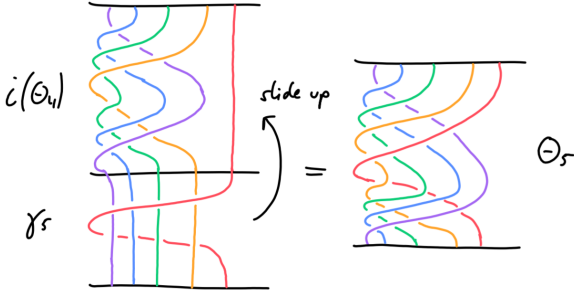


Figure 32: The equality $\theta_n = \iota(\theta_{n-1})\gamma_n$ for $n = 5$.

in fig. 32. Furthermore, note that γ_n commutes with all elements of $\iota(P_{n-1})$, since sliding γ_n up or down does not interfere with any of the crossings in the first $n-1$ strands, see fig. 33. It will turn out, that γ_n generates the U_n -part of a braid in $Z(P_n)$.

Regarding half-twists, we have $\sigma_i \Delta_n = \Delta_n \sigma_{n-i}$ which can be visualized by sliding the crossing up or down, see fig. 34. This implies $\theta_n \in Z(B_n)$, since we have

$$\sigma_i \theta_n = \sigma_i \Delta_n \Delta_n = \Delta_n \sigma_{n-i} \Delta_n = \Delta_n \Delta_n \sigma_i = \theta_n \sigma_i.$$

Lastly, note that the forgetting map $f_n : P_n \rightarrow P_{n-1}$ restricts to $f_n : Z(P_n) \rightarrow Z(P_{n-1})$. In order to see this,

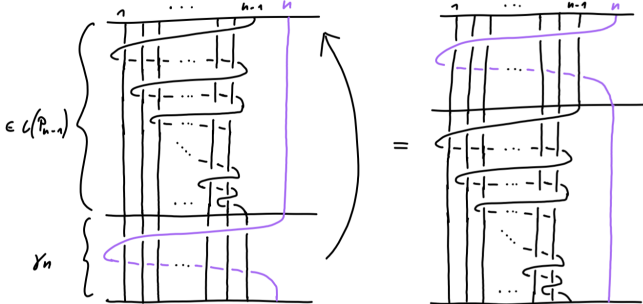


Figure 33: Sliding γ_n up along an element of $\iota(P_{n-1})$.

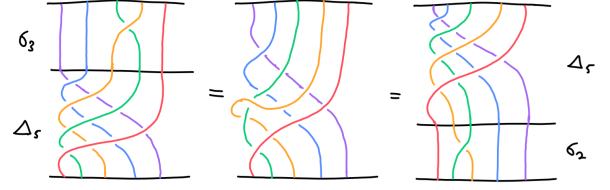


Figure 34: The equality $\sigma_i \Delta_n = \Delta_n \sigma_{n-i}$ for $n = 5$.

let $\beta \in Z(P_n)$, then we want to show that its image $f_n(\beta)$ commutes with arbitrary elements $\alpha \in P_{n-1}$. Indeed, take such an α , then consider $\iota(\alpha) \in P_n$. By assumption, $\beta \iota(\alpha) = \iota(\alpha) \beta$. Using that ι is a right-inverse of f_n , we get

$$\begin{aligned} f_n(\beta) \alpha &= f_n(\beta) f_n(\iota(\alpha)) \\ &= f_n(\beta \iota(\alpha)) \\ &= f_n(\iota(\alpha) \beta) \\ &= \alpha f_n(\beta). \end{aligned}$$

(Note that this is a special case of a more general result about surjective homomorphisms mapping center to center.)

Now we come to one of the two main results of this section. The proof is taken from theorem 1.24 in [CK08]. In the reference, the first statement of the theorem is formulated only for $n \geq 3$, but following the proof, it can be seen that it also holds for $Z(P_2)$.

The rough outline of the proof is as follows:

- i) Show by induction on $n \geq 2$ that $Z(P_n) = \langle \theta_n \rangle$. It needs to be shown that $\beta \in Z(P_n)$ implies $\beta \in \langle \theta_n \rangle$. Here we use the normal form $\beta = \iota(\beta') \beta_n$. The P_{n-1} -part β' can then be written as $(\theta_{n-1})^m$ for some m , and the U_n -part β_n can be written as γ_n^k for some k . The length homomorphism from corollary 4.5 then implies $m = k$. The result follows by using that θ_n is constructed from $\iota(\theta_{n-1})$ and γ_n as discussed above, thus $\beta = \theta_n^m$.
- ii) The fact that $Z(B_n) = Z(P_n)$ for $n \geq 3$ uses the projection $\pi : B_n \rightarrow \Sigma_n$ which maps $Z(B_n) \rightarrow Z(\Sigma_n) = 1$. Using $\theta_n \in Z(B_n)$ as discussed before, one then shows that $Z(B_n) \subseteq Z(P_n) \subseteq Z(B_n)$.

Theorem 4.6. *For $n \geq 2$, the center $Z(P_n) = \langle \theta_n \rangle$ is an infinite cyclic group, where θ_n is the full-twist defined above.*

Furthermore, for $n \geq 3$ we have $Z(B_n) = Z(P_n)$.

Proof. First we show the proof for $Z(P_n) = \langle \theta_n \rangle$ by induction over $n \geq 2$. For $n = 2$, we have $P_2 = \langle A_{1,2} \rangle = \langle \theta_2 \rangle$ by corollary 4.4.

Now assume that the claim is true for $n - 1$. Let

$\beta \in Z(P_n)$. By our discussion about normal forms of pure braids, we know that

$$\beta = \iota(\beta')\beta_n,$$

where $\beta' = f_n(\beta)$ and $\beta_n \in U_n$. We saw earlier that γ_n commutes with all elements of $\iota(P_{n-1})$ and thus

$$\gamma_n \iota(\beta') = \iota(\beta') \gamma_n.$$

We also have

$$\gamma_n \beta = \beta \gamma_n$$

by assumption. Using $\beta_n = (\iota(\beta'))^{-1} \beta$, we obtain

$$\gamma_n \beta_n = \beta_n \gamma_n.$$

Now consider $G = \langle \beta_n, \gamma_n \rangle \subseteq U_n$. By the last equality, this is an abelian group. It is also free as a subgroup of U_n . Hence it must be isomorphic to $F_1 = \mathbb{Z}$. Recall the length homomorphisms $\ell_{i,n} : P_n \rightarrow \mathbb{Z}$ used in the proof of corollary 4.5. Applied to $\gamma_n = A_{1,n} \cdots A_{n-1,n}$ we have $\ell_{1,n}(\gamma_n) = 1$, implying that γ_n is a generator of the cyclic group G , which yields

$$\beta_n = \gamma_n^k$$

for some k .

Next, since f_n restricts to $Z(P_n) \rightarrow Z(P_{n-1})$, by induction assumption we have

$$\beta' = f_n(\beta) = (\theta_{n-1})^m$$

for some m .

All in all, we have $\beta = \iota(\theta_{n-1})^m \gamma_n^k$. Recalling the definition of γ_n , one then obtains

$$\ell_{i,n}(\beta) = k,$$

because $\theta_{n-1} \in P_{n-1}$. Notably, this value does not depend on n . Furthermore this holds for all elements in $Z(P_n)$.

By assumption, $\beta \in Z(P_n)$, thus also

$$\sigma_{n-1} \beta \sigma_{n-1}^{-1} \in Z(P_n),$$

since adding a crossing in the last two strands and then reversing it afterwards does not change how the braid commutes with others.

Now one can compute

$$\ell_{1,n}(\sigma_{n-1} \beta \sigma_{n-1}^{-1}) = \ell_{1,n-1}(\beta) = m$$

and

$$\ell_{n-1,n}(\sigma_{n-1} \beta \sigma_{n-1}^{-1}) = \ell_{n-1,n}(\beta) = k.$$

We have seen that this value does not depend on the first

index and thus $m = k$. Using this, we get

$$\begin{aligned} \beta &= \iota(\beta')\beta_n \\ &= \iota((\theta_{n-1})^m)\gamma_n^k \\ &= \iota((\theta_{n-1})^k)\gamma_n^k \\ &= (\iota(\theta_{n-1})\gamma_n)^k \\ &= (\theta_n)^k, \end{aligned}$$

concluding the induction.

Indeed, $\langle \theta_n \rangle$ is torsion-free as a subgroup of P_n by corollary 4.3, since subgroups of torsion-free groups must be torsion-free.

It remains to show that $Z(B_n) = Z(P_n)$ for $n \geq 3$. Here we can use that surjective homomorphisms map center to center and since $Z(\Sigma_n) = 1$ for $n \geq 3$, we have that the projection $\pi : B_n \rightarrow \Sigma_n$ maps the center to 1, thus $Z(B_n) \subseteq \ker \pi = P_n$, i.e. every element in the center of B_n is a pure braid.

Combined with the fact that every element in the center of B_n commutes in particular with all elements of P_n , we obtain $Z(B_n) \subseteq Z(P_n)$.

Furthermore, in the discussion before this theorem it was remarked that $\theta_n \in Z(B_n)$. All in all, this yields

$$Z(B_n) \subseteq Z(P_n) = \langle \theta_n \rangle \subseteq Z(B_n),$$

showing $Z(B_n) = Z(P_n)$ if $n \geq 3$. $\square$

As a consequence, we easily get the main result of this document. The proof is based on [CK08], corollary 1.25. It builds upon the fact, that for $m \neq n$, the set of integer multiples of the length of θ_n differs from the set of integer multiples of the length of θ_m .

Corollary 4.7. *For $m \neq n$, we have $B_m \not\cong B_n$.*

Proof. Recall proposition 3.8, where we saw that there is a length homomorphism $f : B_n \rightarrow \mathbb{Z}$ which maps every generator σ_i to 1. Furthermore it induces an isomorphism $\bar{f} : B_n^{\text{ab}} \rightarrow \mathbb{Z}$.

The image of the full-twist under the length map is

$$\begin{aligned} f(\theta_n) &= f(\Delta_n^2) \\ &= 2 \cdot f(\Delta_n) \\ &= 2 \cdot f(\sigma_1 \cdots \sigma_{n-1}(\sigma_1 \cdots \sigma_{n-2}) \cdots (\sigma_1 \sigma_2) \sigma_1) \\ &= 2 \cdot ((n-1) + (n-2) + \cdots + 2 + 1) = n(n-1), \end{aligned}$$

so $f(Z(B_n)) = f(\langle \theta_n \rangle) = n(n-1)\mathbb{Z}$. Hence, the image of $Z(B_n)$ in $B_n^{\text{ab}} \cong \mathbb{Z}$ is a subgroup of $\mathbb{Z}$ of index $n(n-1)$. If $B_m \cong B_n$, then $m(m-1) = n(n-1)$ and thus $m = n$. $\square$

5 Conclusion

In this document we explored braid groups from three different perspectives, while considering connections be-

tween the geometrical, algebraic and topological approaches. These relationships could then be utilized to deduce algebraic properties of braid groups.

At this point, one could continue in various directions. The introduction was hinting at applications in topological robotics, but there are also connections to knot theory, for example. We refer the interested reader to [Abr17].

Also a whole new world opens up when considering (unlabeled) configuration spaces $\mathcal{F}_n(M)$ of topological manifolds M instead of $\mathbb{R}^2$, arriving at the notion of a *(pure) braid group of M* when considering the fundamental group $\pi_1(\mathcal{F}_n(M))$, see chapter 1.4 of [CK08]. This has some implications for mapping class groups, for example, see chapter 1.6 of [CK08].

Personally, I hope that the reader is now convinced that braid groups are a versatile and interesting, yet also somewhat playful, object of study.

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